Recent Research in Computer Graphics: Article List

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The following 8 articles were selected from ACM Transactions on Graphics (TOG) and IEEE Transactions on Visualization and Computer Graphics (TVCG), published in 2024-2025. Papers are organized into thematic clusters.

Cluster 1: Procedural Noise and Texturing

[1] A. Maesumi, D. Hu, K. Saripalli, V. G. Kim, M. Fisher, S. Pirk, and D. Ritchie, "One Noise to Rule Them All: Learning a Unified Model of Spatially-Varying Noise Patterns," *ACM Transactions on Graphics*, vol. 43, no. 4, pp. 1-21, Jul. 2024.

Keywords: *procedural noise, texture synthesis, diffusion models, deep generative models*

This paper trains a single denoising diffusion model to generate and blend multiple types of procedural noise (including Perlin, Gabor, and wavelet), producing spatially-varying noise textures with controllable parameters and demonstrating improved results in inverse procedural material design.

[2] P. Guehl, R. Allegre, G. Gilet, B. Sauvage, M.-P. Cani, and J.-M. Dischler, "Multi-Dimensional Procedural Wave Noise," *ACM Transactions on Graphics*, vol. 44, no. 4, pp. 1-15, Aug. 2025.

Keywords: *procedural noise, wave noise, spectral control, real-time shading*

This paper introduces a wave-based procedural noise primitive that naturally extends to arbitrary dimensions, offering fine spectral control and efficient GPU evaluation as an alternative to classical gradient-based noise functions like Perlin noise.

[3] Y. Zhang, H. Xu, Y. Wu, S. Chen, S. Lin, X. Li, X. Gao, and X. Jin, "The Mokume Dataset and Inverse Modeling of Solid Wood Textures," *ACM Transactions on Graphics*, vol. 44, no. 4, 2025.

Keywords: *solid texturing, wood appearance, inverse modeling, material acquisition*

This paper presents a dataset of real wood cross-sections and an inverse procedural modeling pipeline that reconstructs 3D solid wood textures from photographs, directly extending the solid texturing concepts introduced in Perlin's 1985 work.

Cluster 2: Real-Time and Neural Rendering

[4] C. Wang, S. Liu, X. Chen, X. He, Y. Huo, and R. Wang, "LightFormer: Light-oriented Global Neural Rendering in Dynamic Scenes," *ACM Transactions on Graphics*, vol. 43, no. 4, pp. 1-15, Jul. 2024.

Keywords: *global illumination, neural rendering, real-time rendering, dynamic scenes*

This paper proposes a neural rendering approach that represents light sources rather than entire scenes using virtual point lights and a pixel-light attention mechanism, achieving real-time global illumination for fully dynamic scenes with changing lighting, materials, and animated objects.

[5] W. Liu, X. Zheng, X. Li, T. Y. Al-Naffouri, J. Yu, and X. Lou, "CoARF++: Content-Aware Radiance Field Aligning Model Complexity With Scene Intricacy," *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 10, pp. 8187-8199, Oct. 2025.

Keywords: *neural radiance fields, content-aware rendering, model compression, novel view synthesis*

This paper presents an adaptive neural radiance field method that automatically allocates model complexity based on scene detail, achieving efficient rendering by using fewer parameters in simple regions while preserving quality in visually complex areas.

[6] C. Kleinbeck, H. Schieber, K. Engel, R. Gutjahr, and D. Roth, "Multi-Layer Gaussian Splatting for Immersive Anatomy Visualization," *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 5, pp. 2353-2363, May 2025.

Keywords: *Gaussian splatting, medical visualization, virtual reality, volume rendering*

This paper introduces a layered Gaussian splatting representation for CT scan visualization in VR, enabling interactive anatomical exploration (e.g., toggling bone, muscle, and tissue layers) at real-time frame rates on compute-constrained devices like mobile headsets.

Cluster 3: Rendering Efficiency and Optimization

[7] S. Yang, Y. Zhao, Y. Luo, H. Wang, H. Sun, C. Li, B. Cai, and X. Jin, "MNSS: Neural Supersampling Framework for Real-Time Rendering on Mobile Devices," *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 7, pp. 4304-4320, Jul. 2024.

Keywords: *neural supersampling, mobile rendering, real-time performance, anti-aliasing*

This paper proposes a lightweight neural network framework that upsamples low-resolution rendered frames to high resolution in real time on mobile GPUs, addressing the efficiency gap between desktop and mobile graphics hardware for interactive applications.

[8] Y. Zhang, H. Xu, Y. Wu, S. Chen, X. Li, and X. Jin, "Texture Size Reduction Through Symmetric Overlap and Texture Carving," *ACM Transactions on Graphics*, vol. 44, no. 4, 2025.

Keywords: *texture compression, memory optimization, texture mapping, GPU rendering*

This paper presents techniques to reduce texture memory footprint by exploiting symmetric overlap and carving away unused regions of texture atlases, directly addressing the same memory efficiency concerns that motivated Perlin's original move from stored images to procedural textures.